MatriWatch

Smart Material Health Monitoring Software For Mothers

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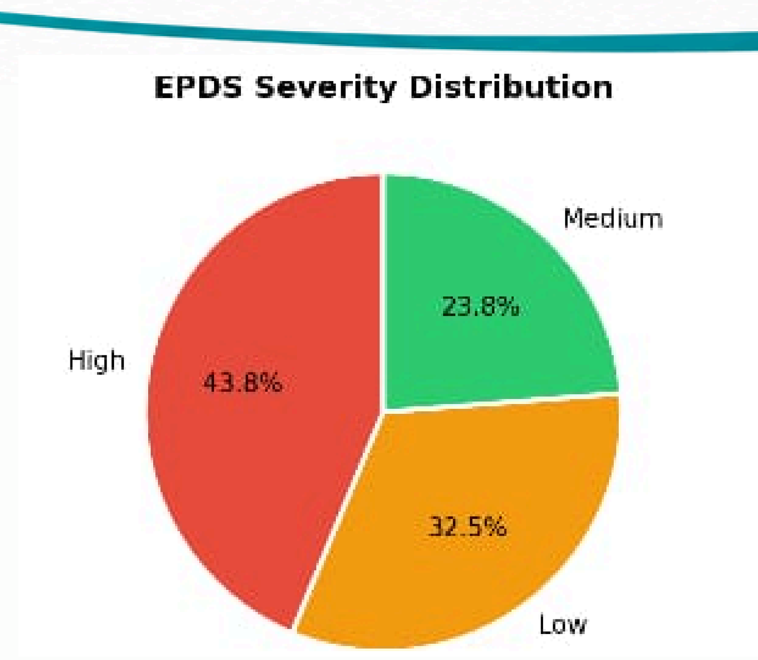
Team MatriWatch

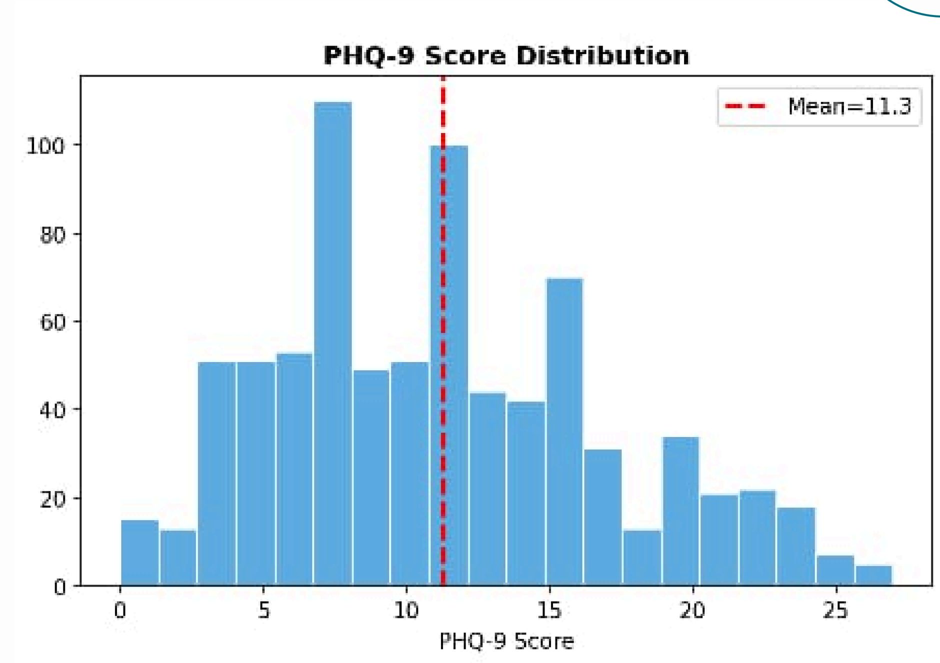
1 LITERATURE REVIEW

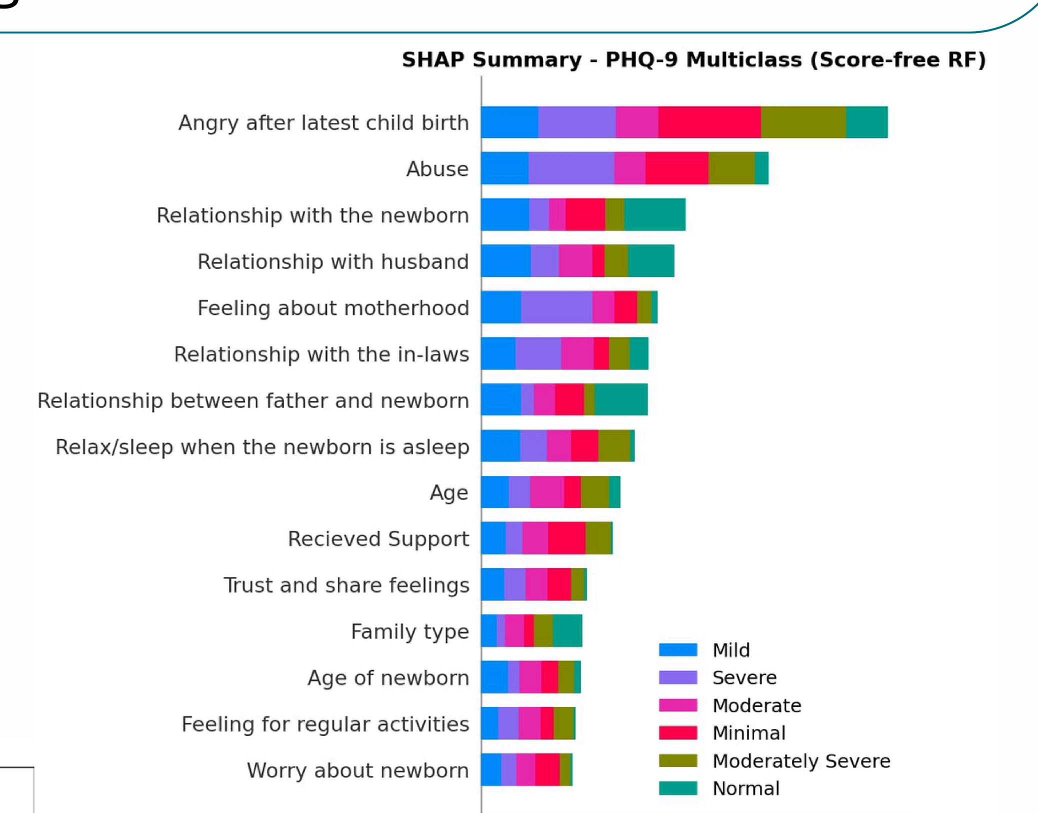
Postpartum depression affects mothers, infants, and family well-being, but early detection is still limited in low-resource settings. Tools like PHQ-9 and EPDS are widely used, while machine learning can improve fast and data-driven screening.



3 RESEARCH GAP

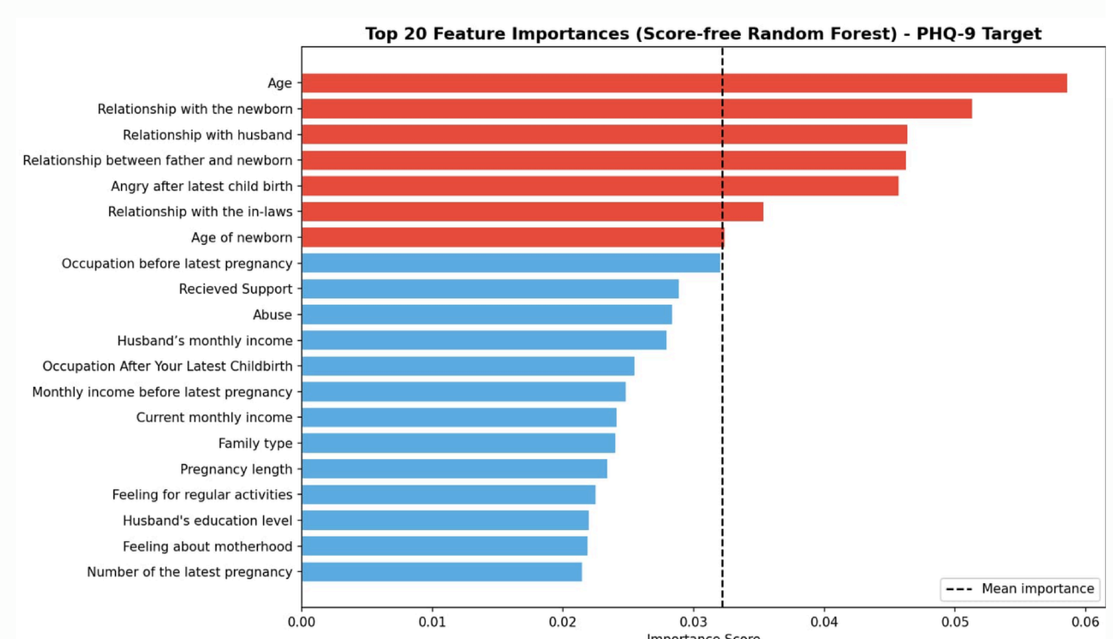
Most studies focus on binary detection or depend on raw depression scores. Limited work explores fine-grained PHQ-9 and EPDS classification using Bangladesh-focused postpartum data with explainable machine learning.

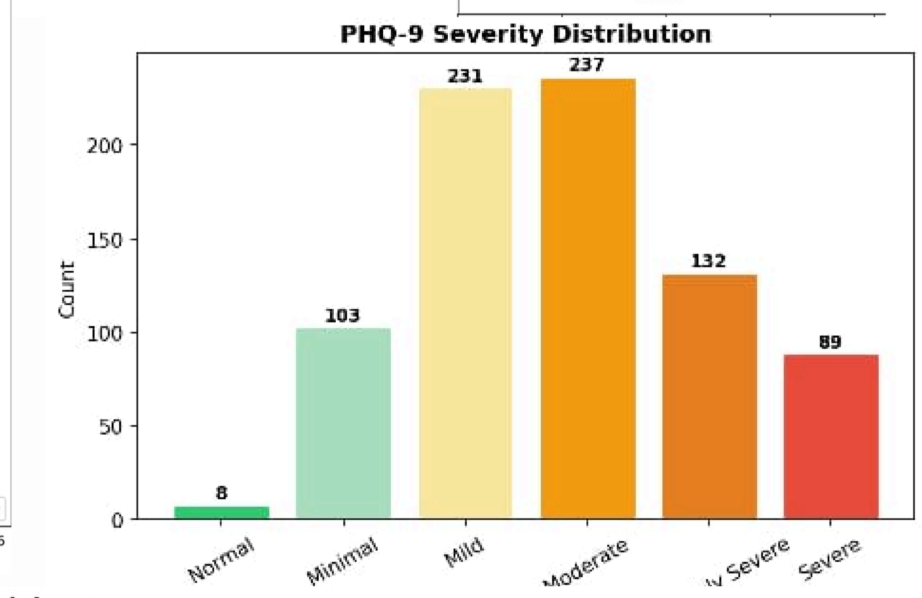




2 RESEARCH OBJECTIVE

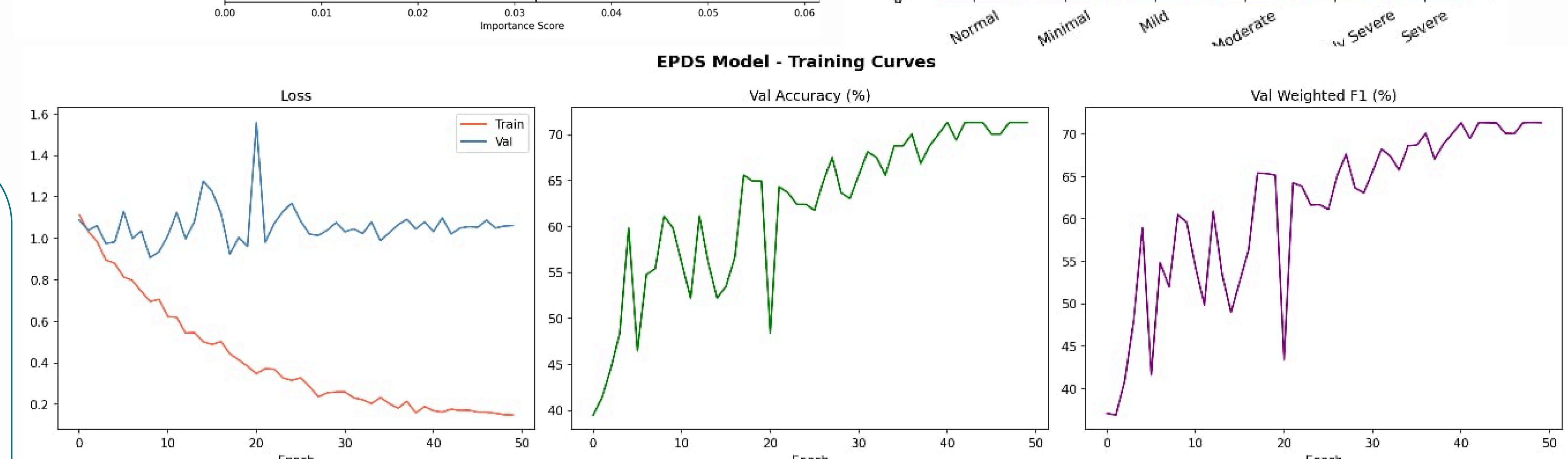
- Build a Bangladesh-focused postpartum depression screening model.
- Predict PHQ-9 severity and EPDS risk level.
- Compare Attention-based 1D-CNN and RBF-SVM.
- Handle missing values and class imbalance.
- Explain predictions using SHAP and feature importance.

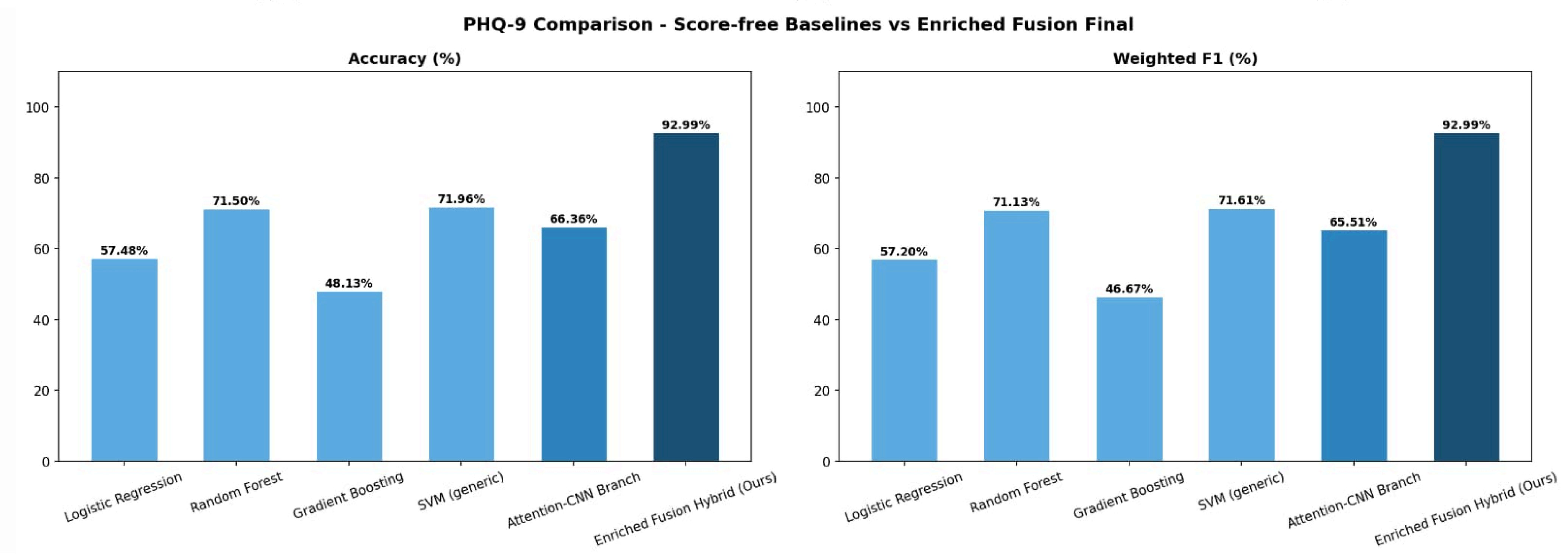


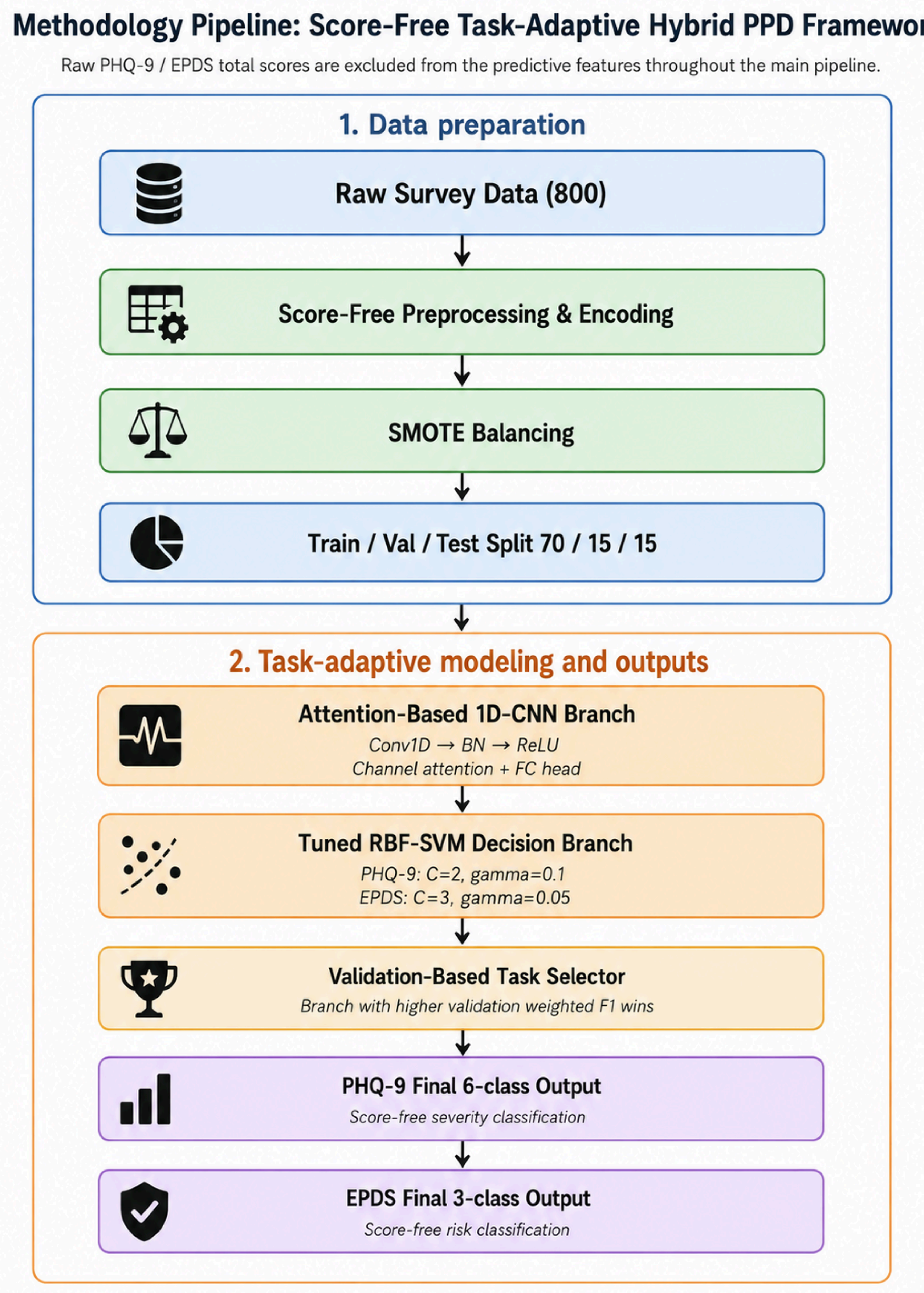


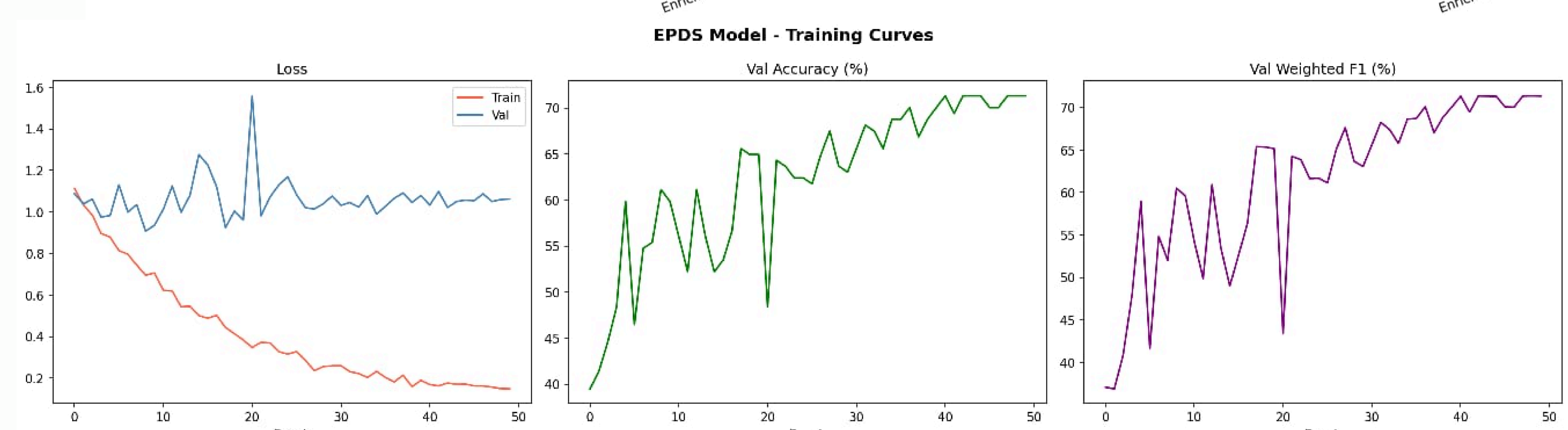
4 METHODOLOGY

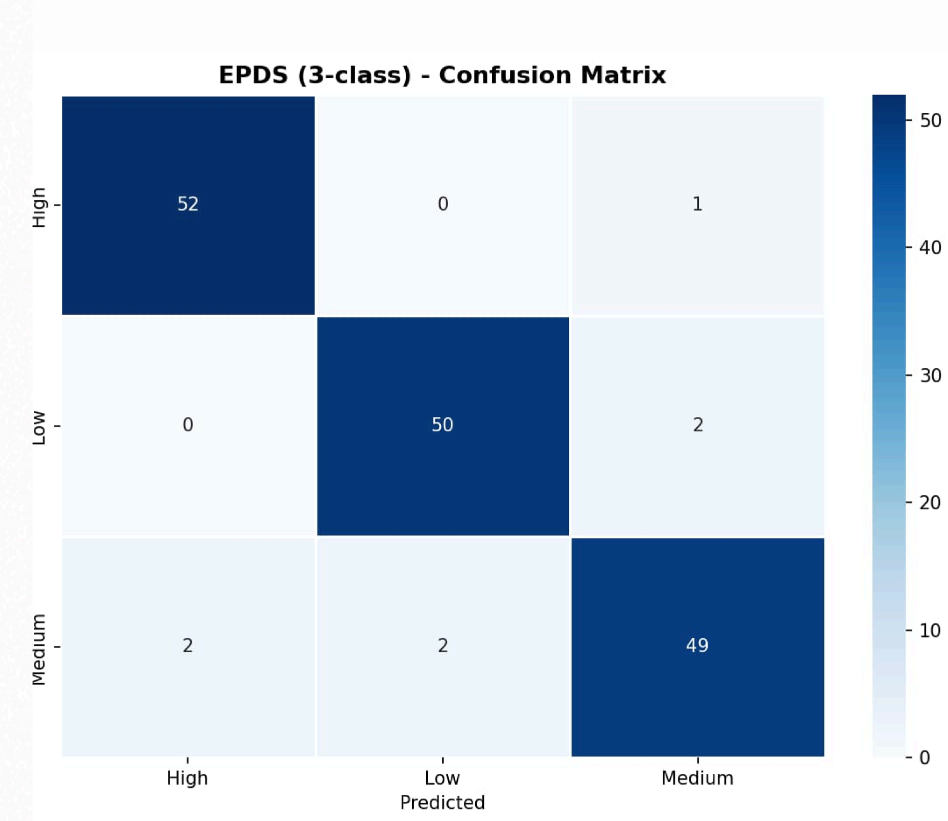
The dataset of 800 postpartum women was cleaned, encoded, standardized, and balanced using SMOTE. Two models, Attention-based 1D-CNN and RBF-SVM, were trained and compared. The best model was selected using validation performance and tested using accuracy, weighted F1-score, ROC curves, confusion matrix, and SHAP analysis.











5 CONCLUSION

The proposed score-free model achieved strong performance for both PHQ-9 and EPDS screening. Results show that psychosocial, relational, abuse-related, newborn, and postpartum factors can support early and interpretable depression risk detection in low-resource maternal healthcare settings.